

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	An urn A contains 2 white and 4 black balls. Another urn B contains 5 white and 7 black balls. A ball is transferred from urn A to urn B. If a ball is drawn from urn B, find the probability that it is white.	5	CO1
1b	A Quantizer of a digital communication system producing the quantization error Q will have its sample values uniformly distributed and bounded by $-\Delta/2 \leq q \leq \Delta/2$ where Δ is the step size of the quantizer. Find the mean square error of the quantizer. Suggest the way to minimize the quantization error.	5	CO1
1c	Describe Gaussian distribution. Derive its mean and variance.	5	CO1
Q 2	Attempt any 2 parts of the following		
2a	Explain joint distribution and joint density function with example.	5	CO2
2b	The joint probability density function of two random variables X and Y is given by $f_{XY}(x, y) = C(2x + y), \quad 0 \leq x \leq 2, \quad 0 \leq y \leq 3$ $= 0, \text{ otherwise}$ Find the Value of C Also Find $P[X \geq 1, Y \leq 2]$	5	CO2
2c	Explain marginal distribution, conditional distribution and conditional expectation.	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	A random variable X has a mean of 8, variance of 9 and an unknown probability distribution function. Find $P[-4 < X < 20]$ using Chebyshev's Inequality. What conclusion you can make from this inequality theorem.	5	CO3
3b	X is a random variable with the PDF given by $f_X(x)$, and $Y = g(X)$. Derive the PDF of Y .	5	CO3
3c	The PDF of a random variable X is given by $f_X(x) = 1$, for $0 < x \leq 1$, and zero elsewhere. Determine the PDF of Y where $Y = 4X + 5$.	5	CO3
Q 4	Attempt any 2 parts of the following		
4a	Explain Wide Sense Stationary (WSS) process and its properties.	5	CO4
4b	Given a stationary random process $X(t) = 10 \cos[100t + \phi]$, where ϕ is a random variable with a uniform probability distribution in the interval $[-\pi, \pi]$. Show that $X(t)$ is ergodic in terms of Auto correlation function.	5	CO4
4c	The autocorrelation function of a random process is given by, $R_{xx}(\tau) = 36 + \frac{9}{1+6\tau^2}$. Find mean and variance of the process.	5	CO4
Q 5	Attempt any 2 parts of the following		
5a	State and prove three significant properties of Power Spectral Density	5	CO5
5b	If a WSS random process is applied as input to an LTI system, show that the output will be a WSS random process.	5	CO5
5c	Explain the significance of white Noise. Draw the plots for i) Power Spectral Density of White Noise ii) Auto correlation function of White Noise	5	CO5